

Ductwork Layout Drawing & Specifications

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Project: Lakefront Office Tower — Phase II

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SECTION 23 31 13 — METAL DUCTS

1.0 DUCT CONSTRUCTION STANDARDS

All ductwork per SMACNA HVAC Duct Construction Standards 3rd Edition. Pressure class per schedule below. Sheet metal gauges per SMACNA Table 1-1.

2.0 MATERIAL & INSULATION SCHEDULE

Duct System	Pressure Class	Material	Gauge	External Insulation
Supply Main	2-in. SP	G90 Galv. Steel	20 ga.	2-in. duct wrap R-8
Supply Branch	2-in. SP	G90 Galv. Steel	22 ga.	1.5-in. duct wrap R-6
Return Main	1-in. SP	G90 Galv. Steel	22 ga.	None (plenum)
Return Branch	1-in. SP	G90 Galv. Steel	24 ga.	None
Exhaust	1-in. SP	G90 Galv. Steel	24 ga.	None
Kitchen Exhaust	2-in. SP	304 SS, 16 ga.	16 ga.	1-hr fire-wrap
OA / Relief	1-in. SP	G90 Galv. Steel	22 ga.	2-in. R-8 (exterior)

3.0 TYPICAL FLOOR PLAN — DUCT LAYOUT (FLOOR 5)

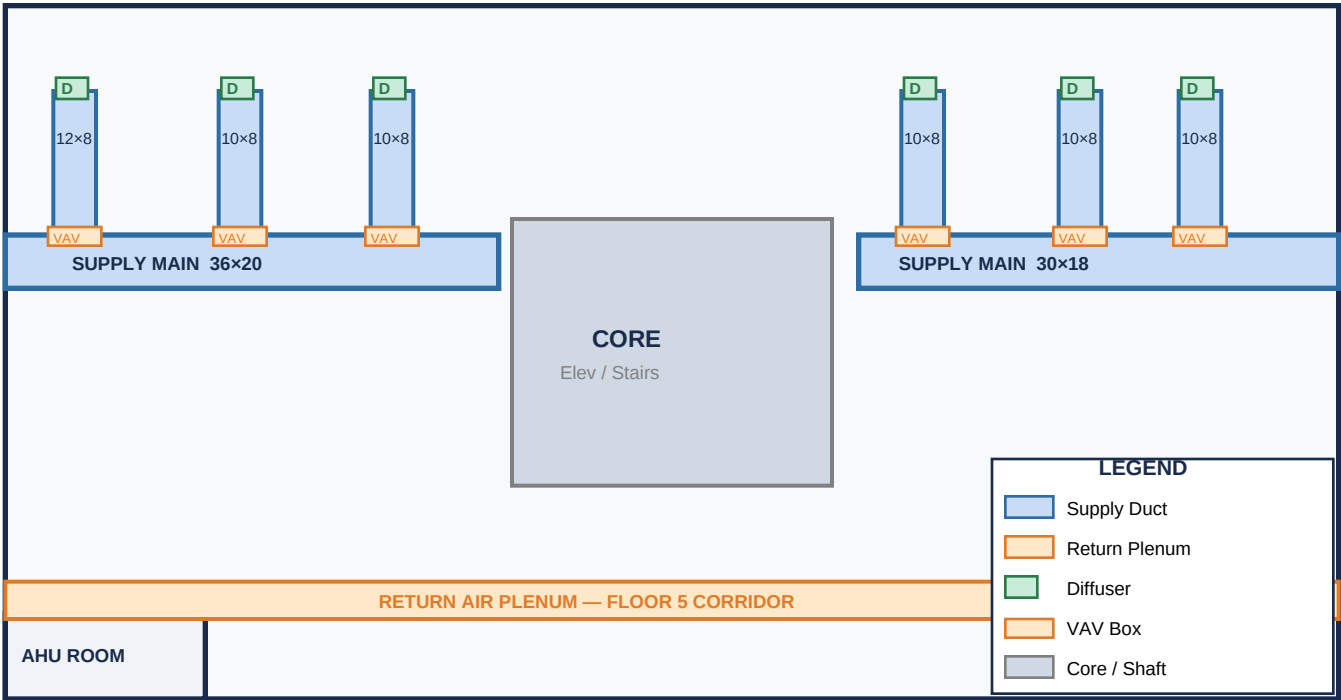


Figure 2 — Typical Floor Duct Layout, Floor 5 (Schematic — Not to Scale)

4.0 VAV BOX SCHEDULE

Tag	Zone	CFM Max	CFM Min	Box Size (in.)	Coil (MBH)	Ctrl
VAV-501	Open Office E	2,400	480	12×12	12	DDC
VAV-502	Open Office W	2,200	440	12×12	12	DDC
VAV-503	Conference 501	800	200	8×8	6	DDC
VAV-504	Conference 502	600	160	8×8	4	DDC
VAV-505	Private Off. A	350	90	6×6	3	DDC
VAV-506	Private Off. B	350	90	6×6	3	DDC
VAV-507	Break Room	500	150	8×6	4	DDC
VAV-508	Reception	400	100	6×6	3	DDC

5.0 INSTALLATION NOTES

- All duct joints sealed with UL 181 mastic or foil-faced tape rated for pressure class.
- Flexible duct connections: 18 inches maximum length.
- Rectangular duct >19-in. width: intermediate hangers at 4-ft max spacing.
- Access doors min 12×12 in. at all fire/smoke dampers and coil valves.
- Leakage testing for pressure class 2 systems per SMACNA Leakage Class 6.
- Vapor barrier facing must be continuous and un-punctured.

END OF SECTION 23 31 13